# PROTON-QA 1.0: A Nucleus for Provenance-Aware Ontology-Grounded Nuclear QA Benchmark

Thushari Pahalage Dona[1], Antonio Bulgheroni[2], Lorenzo Fongaro[2] and Matteo Lissandrini[1]

[1]Department of Computer Science, University of Verona, Verona, Italy
[2]European Commission, Joint Research Centre, Karlsruhe, Germany

**Abstract**

Retrieval-based Question Answering (QA) systems retrieve information from documents to generate answers to user questions. Existing approaches usually generate plain-text answers, omitting important annotations related to provenance and domain knowledge which are critical in highly regulated technical applications. In technical domains, answer validity depends on correct provenance, unit-aware interpretation of numerical values, measurement uncertainty, and interpretation of specialized terminology. These requirements are especially important when evaluating Retrieval-Augmented Generation (RAG) systems using Large Language Models (LLMs). In this context, plain text is not sufficient; instead, QA systems should produce semantically annotated, verifiable, and computable answers. In this work, we present PROTON-QA 1.0, a provenance-based, ontology-linked QA benchmark derived from archival nuclear domain reports and expert questions. The benchmark serves as a nucleus for future QA benchmarks in scientific domains by laying the semantic and provenance-aware foundation for: (1) document-level provenance traceability, (2) unit-aware numerical answer representation, (3) concept-level entity linking to external knowledge resources, and (4) ontology-grounded RDF representation of questions and answers. The current release contains 50 expert-curated questions derived from nuclear-domain reports spanning 14 years totaling about 1000 pages, enriched with fine-grained provenance, QUDT-based numerical annotations, named-entity mappings to Wikidata and EuroVoc, and custom concept extensions for domain-specific terms not represented in existing ontologies.

**Keywords**

Knowledge Representation, Provenance, Ontology Grounding, Question Answering, Nuclear Data Management

## 1. Introduction & Related Work

Recent advances in retrieval-augmented generation (RAG), Large Language Models (LLMs), and Knowledge Graph Question Answering (KGQA) have provided new state-of-the-art performance for reliable document-based question answering [1]. However, particularly in highly regulated safety-critical domains, answer quality depends not only on retrieval and generation performance but also on the ability to verify answers against factual source evidence [2]. Consequently, reliable QA systems require traceable provenance linking answers to supporting evidence [3]. This is in contrast with current practice, where most systems produce plain text as answers with minimal annotations w.r.t. the provenance of their answers, and similarly, existing benchmarks do not explicitly evaluate this aspect. Technical domains further introduce challenges often overlooked by QA benchmarks, including unit-aware numerical values, measurement uncertainty, specialized terminology, and context-dependent answers [4].

In this work, we focus on the nuclear domain as a reference application area for developing a foundational provenance-aware and ontology-grounded QA benchmark motivated by a nuclear waste reporting use case developed with the European Commission's Joint Research Centre (JRC), Karlsruhe, where AI-assisted document analysis requires rigorous evaluation before deployment in high-stakes settings. Nuclear knowledge spans diverse subdomains including nuclear waste management, nuclear medicine, radiochemistry, reactor engineering, and fuel cycle research, where terminology, measurements, and scientific concepts often vary across application contexts and experimental settings [5]. The above

*ISWC 2026 Companion Volume, October 25–29, 2026, Bari, Italy*
thushari.pahalage@univr.it (T. P. Dona); Antonio.BULGHERONI@ec.europa.eu (A. Bulgheroni); Lorenzo.FONGARO@ec.europa.eu (L. Fongaro); matteo.lissandrini@univr.it (M. Lissandrini)
0000-0001-6563-9516 (T. P. Dona); 0000-0001-7988-2090 (A. Bulgheroni); 0000-0001-9267-3525 (L. Fongaro); 0000-0001-7922-5998 (M. Lissandrini)
© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

**Table 1**
Comparison of QA resources across semantic-web and scientific knowledge representation dimensions.

| Dataset | Domain | Knowledge Source | #Qs | Representation | Ontology Grounding | Retrieval Scope | Provenance | Unit-Aware | Expert-Validated |
|---|---|---|---|---|---|---|---|---|---|
| SQuAD [6] | Open-domain | Wikipedia | 107,785 | JSON | None | Single-doc | Paragraph | No | No |
| TriviaQA [7] | Open-domain | Web & Wikipedia | 95,956 | JSON | None | Multi-doc | Supporting documents | No | No |
| QALD-9-plus [8] | Open-domain | DBpedia, Wikidata | 558* | JSON | DBpedia, Wikidata | KG | KG triple(s)¶ | Limited | Partial§ |
| ACORD [9] | Legal | Legal documents | 114† | JSONL/TSV | None | Single-doc | Document | No | Yes |
| PubMedQA [10] | Biomedical | Abstracts | 1,000 | JSON | None | Single-doc | Abstract | No | Yes |
| SPIQA [11] | Scientific | Research articles | 270,000 | JSON | None | Single-doc | Text, figures & tables | No | Partial§ |
| MDA-QA [12] | Scientific | Research articles | 6,804 | JSON | None | Multi-doc | Supporting articles | No | Partial |
| PeerQA [13] | Scientific | Research articles | 579 | JSON | None | Single-doc | Evidence spans | No | Yes |
| SciDQA [14] | Computer Sci. | Research articles + reviews | 2,937 | JSON | None | Single-doc[a] | Text + reviews + responses | No | Yes |
| M3SciQA [15] | Computer Sci. | Research articles | 1,452 | JSON | None | Multi-doc | Paper cluster | No | Yes |
| NuclearQA [16]‡ | Nuclear | None | 100 | CSV | None | None | None | No | Yes |
| NQuAD [17]‡ | Nuclear | Technical reports | 612 | JSON | None | Single-doc | Passage | No | Yes |
| **PROTON-QA** | **Nuclear** | **Technical reports** | **50** | **RDF** | **Wikidata, QUDT, EuroVoc, custom** | **Corpus** | **Page + text span** | **Yes** | **Yes** |

*507/558 questions include valid Wikidata queries; all include DBpedia queries.
¶SPARQL queries and returned bindings serve as provenance. §Manual, non-expert validation. †114 queries, 126,659 query–clause judgments. [a]Approximately 11% require multi-document reasoning. ‡Available only as an arXiv preprint.

characteristics make the nuclear domain particularly suitable for studying provenance-aware and semantically grounded QA systems. We reviewed existing QA benchmarks covering general, scientific, and nuclear domains to determine whether they provide the representation and annotation capabilities required for this use case.

As summarized in Table 1, existing resources address individual or selected aspects of the problem, such as document retrieval, evidence spans, expert validation, or scientific documents, but generally do not provide a unified representation of domain concepts, quantitative measurements and units, alternative reported values, and document- and passage-level provenance. The general-domain benchmarks primarily evaluate textual answer extraction, whereas the scientific resources extend these tasks to technical documents and, in some cases, evidence spans, figures, tables, or multi-document settings. However, these resources address different aspects of scientific QA and do not provide the full combination of domain concepts and relationships, measurement information, ontology grounding, and provenance annotations needed to interpret reported values in their scientific context and evaluate their supporting evidence. Existing nuclear-domain benchmarks similarly target technical documents, yet provide only limited support for provenance-aware evaluation, semantic grounding, and unit-aware numerical representations. Moreover, document-based benchmarks generally assume that the relevant evidence has already been identified, evaluating question answering over a predefined paragraph, document, or pre-filtered corpus rather than evaluating the complete retrieval-and-grounding process. The identified limitations are particularly relevant for nuclear technical reports, where the scientific interpretation of an answer depends on its measurement context, domain terminology, and supporting documentary evidence. These requirements motivated the design of a structured benchmark in which these elements are represented explicitly and can be evaluated together.

We present PROTON-QA 1.0, a nucleus and reference resource for provenance-aware and ontology-grounded benchmarking of QA systems in technical domains. We propose modeling the benchmark data and annotations in the Resource Description Framework (RDF). To ensure interoperability, PROTON-QA is represented using established Semantic Web vocabularies. The benchmark builds upon Schema.org [18], QAnary [19], and Open Annotation (OA) [20] for representing questions and answers. In particular, it adopts the annotation principles of OA while extending them with provenance-aware and ontology-grounded semantics, QUDT [21] for unit-aware quantitative values, and Wikidata [22] and EuroVoc [23] for semantic grounding. Where existing resources lack coverage of specialized nuclear concepts, PROTON-QA incorporates expert-defined semantic extensions to preserve domain-specific meaning. Hence, central to our contribution is the first proposal for a semantic representation for question-answer benchmark pairs tailored to technical-domain QA. We release the data and a prelim-

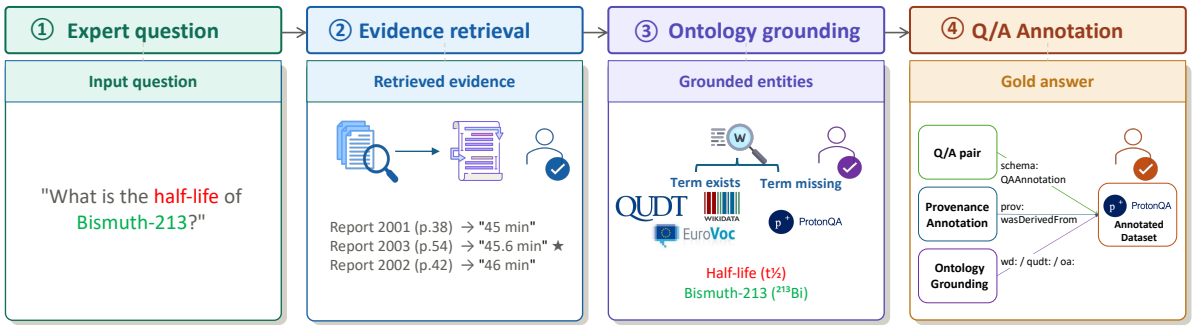



**Figure 1:** Expert-validated grounding workflow from expert question to semantically annotated QA pair.

inary evaluation scripts. The PROTON-QA benchmark and its associated resources are released on GitHub [1], while the source document corpus is also archived separately on Zenodo [24] to provide an independent, citable collection of the underlying technical reports.

## 2. Dataset Construction and Annotation

Nuclear scientists often need to retrieve experimental values and their associated units from archival technical reports. A recent common practice in technical domains is to upload document collections to an LLM-powered question answering systems and interact with them using natural language queries. While these systems simplify access to technical information, evaluating the reliability, provenance, and semantic correctness of their responses remains challenging. PROTON-QA 1.0 is constructed based on these requirements via four steps (see Figure 1). We collected 12 publicly available annual reports published across 14 years between 2000 and 2013 by the JRC (DG JRC, Directorate for Nuclear Safety and Security - Karlsruhe, DE) [24]. The reports capture experimentally validated measurements, technical analyses, and domain knowledge across multiple nuclear subdomains. Each document is assigned a stable identifier `type_YYYY_sequence` that is used throughout the benchmark. Each report has between 35 and 172 pages, for a total of about 1000 pages across all documents. The benchmark consists of 50 questions formulated by experts working with these reports for which an answer can be found in one or more of these reports. Usually, these questions aim at retrieving specific numerical values produced by past analysis or experiments, e.g., "What is the half-life of $^{213}$Bi?". Since multiple reports may contain different measurements of the same quantity, PROTON-QA distinguishes suggested and accepted answers. The accepted answers correspond to the value selected as the reference by domain experts, typically the most recent and precise measurement, while the alternative reported values that are still valid are retained as suggested answers.

To design PROTON-QA we faced few important questions, namely: (1) how to semantically represent question-answer-document tuples, and (2) how to model alternative answers. Alternative answers may exist in one or more documents and they could be all acceptable in theory but in practice the expert has to pick the best option among them to insert in their report along with the annotation of which documents provide supporting evidence for that answer. For example, as illustrated in Figure 1, earlier reports approximate the half-life of $^{213}$Bi as 45 min and 46 min, whereas a later report records the more precise value of 45.6 min. The benchmark preserves all reported evidence but adopts the expert-validated value as the accepted answer, enabling evaluation of multi-document retrieval, provenance verification, and knowledge-grounded reasoning. Hence, each answer is annotated with a link to the page where it appears and answers are modeled as either suggested or accepted answers. Unlike existing benchmarks, PROTON-QA further combines expert-derived ground truth with structured semantic annotations. Hence, we model domain-specific terms and entities in questions and answers by mapping them to ontology concepts, enabling precise semantic interpretation (i.e., grounding). The ontology

[1] https://github.com/ARMADA-DN/Proton-QA

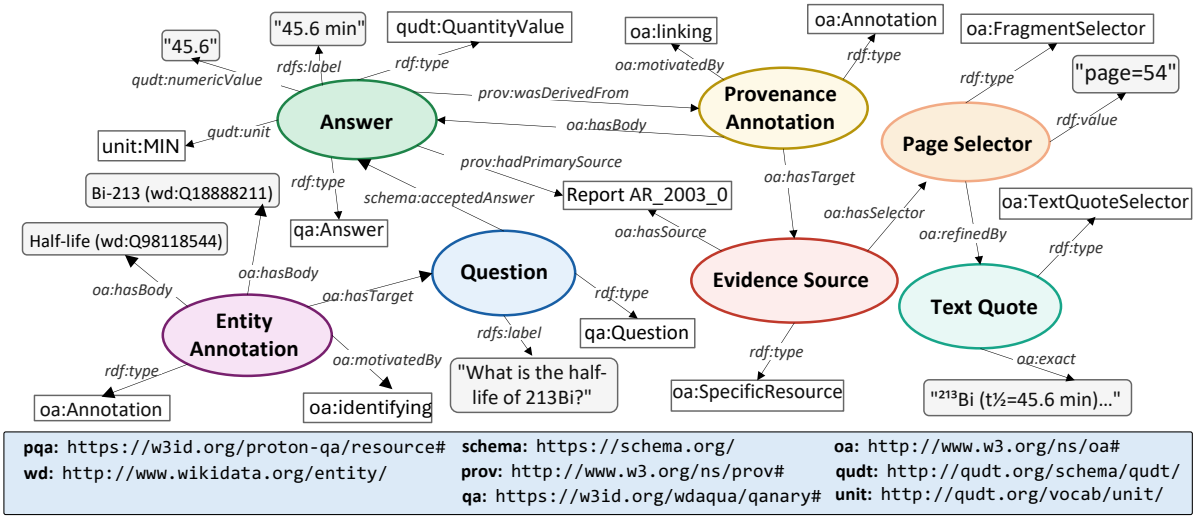



**Figure 2:** PROTON-QA RDF model illustrating provenance and ontology grounding. Document-level provenance identifies the source report, while evidence-level provenance links the accepted answer to the supporting report, page, and evidence quote. Entity annotations ground question concepts.

grounding was performed during the expert annotation process. Its primary purpose is to provide a controlled vocabulary for scientific concepts and to reduce the ambiguity introduced by lexical variation in technical documents. Terms referring to the same scientific entity or concept may appear in different forms, including synonyms, abbreviations, or domain-specific formulations. Mapping these mentions to a common ontology concept provides a consistent semantic reference regardless of the wording used in the source document. Where a suitable existing concept cannot be identified, a locally defined concept is used, as described in Section 3. Further, we provide structured semantic representations of numerical answers, including their units of measurement and, where applicable, uncertainty ranges. This structured format is designed so that future evaluation efforts can validate LLM-generated answers deterministically, rather than relying only on free-text comparison.

## 3. Resource Description & Analysis

Figure 2 illustrates the semantic representation of PROTON-QA. Here, `prov:hadPrimarySource` captures document-level provenance by identifying the source document(s) associated with the QA pair, while `prov:wasDerivedFrom` captures evidence-level provenance by linking the accepted answer to one or more provenance annotations that identify the exact supporting evidence (report, page, and evidence quote). We use QAnary [19] as the core ontology to represent questions and answers, extending it with `schema.org` to associate questions with accepted and alternative answers. OA and PROV-O model provenance annotations, while QUDT [21] represents quantitative values. Entity mentions are linked to Wikidata and, where available, EuroVoc, otherwise, domain-specific concepts are defined in a custom vocabulary using SKOS [25]. The benchmark is distributed through a central `questions.ttl` file containing questions, accepted answers, candidate answers, provenance annotations, and source document references, while the custom concept extension vocabulary is provided in `concept_extensions.ttl`. These concepts are semantically linked to external entities using SKOS mapping relations (e.g., `exactMatch`, `closeMatch`, and `relatedMatch`).

**Preliminary Evaluation.** We conducted a zero-shot retrieval experiment using NotebookLM [26] to evaluate provenance retrieval by comparing the generated responses and retrieved evidence with the corresponding ground truth provenance. We used a predefined prompt to retrieve the answer, source, page, and supporting evidence according to our provenance RDF model in JSON format and the output was then compared with the ground-truth provenance using deterministic metrics. These

provenance-based metrics are complementary to established RAG evaluation approaches such as RAGAS [27] and ARES [28] which assess dimensions including context relevance, context recall, and answer faithfulness. While such approaches can use LLMs as judges to evaluate the relevance and use of retrieved context, we utilize deterministic metrics that directly compare system-retrieved documents, pages, and evidence passages with ground truth provenance, providing an explicit reference for evaluating evidence localization at multiple granularities. In particular, the experiment focuses on whether a system can identify the source documents and supporting locations associated with a benchmark answer. We assessed not only answer correctness but also whether the cited evidence matched the gold-standard provenance. The system correctly identified the right source document in most cases (document recall $\mathrm{DocR} = 0.93$, document precision $\mathrm{DocP} = 0.84$), showing that it can reliably locate which report contains the relevant information. However, this accuracy dropped substantially when we examined *where* in the document the evidence came from: only 40% of the correct supporting pages were retrieved ($\mathrm{PgR} = 0.40$), and just 20% of the cited pages were actually relevant ($\mathrm{PgP} = 0.20$), with the exact supporting passage matching the gold evidence in only 40% of cases ($\mathrm{Evid} = 0.40$). This pattern indicates that while the system is proficient at coarse-grained retrieval, it struggles to pinpoint the precise passage that substantiates its answer; a gap that underscores the value of fine-grained, page- and quote-level provenance annotation for rigorously evaluating QA systems. The benchmark, evaluation scripts, and prompts are publicly available in the benchmark repository.

## 4. Discussion and Future Work

AI-assisted question answering over technical reports is being explored through LLM-based RAG systems that support scientists in retrieving precise information for reports and documentation. In this context, the scientist must be able to verify where an answer originated and how it was interpreted. Thus, in this work we stress the need for semantically annotated benchmarks that allow evaluating a system's ability to provide more than correct answers, and in particular provenance traceability and semantic grounding. By linking questions, answers, domain concepts, quantitative information, and evidence, PROTON-QA provides structured reference annotations for evaluating QA and RAG systems. While the current release is limited by its size, it provides an important step forward as it establishes a rich annotation framework to build benchmarks in this domain. The relative contributions of model capabilities, context limitations, and retrieval strategy require controlled comparison across systems and are therefore left for future evaluation. Our current evaluation is preliminary and based on a single QA system; therefore, the reported results should be interpreted as a demonstration of the evaluation capabilities provided by the benchmark rather than as a comparative assessment of LLM performance. We also plan to extend the corpus and concept vocabulary, incorporate multimodal evidence (e.g., tables and figures), improve automated semantic annotation, and enrich the benchmark with automatic evaluation scripts and metric calculations. Further, we plan to evaluate ontology grounding by comparing the concepts identified by QA systems with the expert-validated ontology mappings in PROTON-QA, serving as the ground truth.

## Acknowledgments

This work is supported by the ARMADA project under the EU Horizon Europe MSCA project-Grant Agreement No. 101168951. Views and opinions are those of the authors and do not necessarily reflect those of the European Union or the European Commission. The European Union and the granting authority are not responsible for them. The authors thank the JRC, Karlsruhe, for its collaboration and for providing the reports used to compile the benchmark.

## Declaration on Generative AI

During the preparation of this work, the author(s) used Grammarly and ChatGPT for grammar, rephrasing, and spelling checks. The authors reviewed and edited all outputs as necessary and take full responsibility for the final content.

## References


[1] J. Chen, H. Zhang, S. Yun, A. Mottini, R. Ying, X. Song, V. Ioannidis, Z. Li, Q. Cui, Gril: Knowledge graph retrieval-integrated learning with large language models, Association for Computational Linguistics, 2025.

[2] G. Kasneci, E. Kasneci, The safety failures we are not instrumenting: a perspective on hidden safety-critical challenges in modern ai systems, AI and Ethics 6 (2026) 295.

[3] S. Amer-Yahia, J. Bogojeska, R. Facchinetti, V. Franceschi, A. Gionis, K. Hose, G. Koutrika, R. Kouyos, M. Lissandrini, S. Maniu, K. Mirylenka, D. Mottin, T. Palpanas, M. Rigotti, Y. Velegrakis, Towards Reliable Conversational Data Analytics, in: 2025 EDBT/ICDT 2025 Joint Conference, Barcelona, Spain, 2025. URL: https://hal.science/hal-05335738.

[4] S. O. Manteaw, Redefining nuclear safety using an integrative concentric paradigm for effective nuclear power regulation, U. Ghana LJ 28 (2015) 42.

[5] K. Kolos, V. Sobes, R. Vogt, C. E. Romano, M. S. Smith, L. A. Bernstein, D. A. Brown, M. T. Burkey, Y. Danon, M. A. Elsawi, et al., Current nuclear data needs for applications, Physical Review Research 4 (2022) 021001.

[6] P. Rajpurkar, J. Zhang, K. Lopyrev, P. Liang, Squad: 100,000+ questions for machine comprehension of text, in: Proceedings of the 2016 conference on empirical methods in natural language processing, 2016, pp. 2383–2392.

[7] M. Joshi, E. Choi, D. S. Weld, L. Zettlemoyer, Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension, in: Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2017, pp. 1601–1611.

[8] A. Perevalov, D. Diefenbach, R. Usbeck, A. Both, Qald-9-plus: A multilingual dataset for question answering over dbpedia and wikidata translated by native speakers, 2022. doi:10.1109/ICSC52841.2022.00045.

[9] S. H. Wang, M. Zubkov, K. Fan, S. Harrell, Y. Sun, W. Chen, A. Plesner, R. Wattenhofer, ACORD: An expert-annotated retrieval dataset for legal contract drafting, 2025. URL: https://aclanthology.org/2025.acl-long.1206/. doi:10.18653/v1/2025.acl-long.1206.

[10] Q. Jin, B. Dhingra, Z. Liu, W. Cohen, X. Lu, Pubmedqa: A dataset for biomedical research question answering, in: Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP), 2019, pp. 2567–2577.

[11] S. Pramanick, R. Chellappa, S. Venugopalan, Spiqa: a dataset for multimodal question answering on scientific papers, 2024.

[12] H. Huang, J. Velcin, Y. Kessaci, Towards multi-document question answering in scientific literature: Pipeline, dataset, and evaluation, in: Findings of the Association for Computational Linguistics: EMNLP 2025, Association for Computational Linguistics, Suzhou, China, 2025, pp. 10867–10881. URL: https://aclanthology.org/2025.findings-emnlp.576/. doi:10.18653/v1/2025.findings-emnlp.576.

[13] T. Baumgärtner, T. Briscoe, I. Gurevych, PeerQA: A scientific question answering dataset from peer reviews, in: Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), Association for Computational Linguistics, Albuquerque, New Mexico, 2025, pp. 508–544. URL: https://aclanthology.org/2025.naacl-long.22/. doi:10.18653/v1/2025.naacl-long.22.

[14] S. Singh, N. Sarkar, A. Cohan, SciDQA: A deep reading comprehension dataset over scientific

papers, in: Y. Al-Onaizan, M. Bansal, Y.-N. Chen (Eds.), Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Miami, Florida, USA, 2024, pp. 20908–20923. URL: https://aclanthology.org/2024.emnlp-main.1163/. doi:10.18653/v1/2024.emnlp-main.1163.

[15] C. Li, Z. Shangguan, Y. Zhao, D. Li, Y. Liu, A. Cohan, M3SciQA: A multi-modal multi-document scientific QA benchmark for evaluating foundation models, in: Findings of the Association for Computational Linguistics: EMNLP 2024, Association for Computational Linguistics, Miami, Florida, USA, 2024, pp. 15419–15446. URL: https://aclanthology.org/2024.findings-emnlp.904/. doi:10.18653/v1/2024.findings-emnlp.904.

[16] A. Acharya, S. Munikoti, A. Hellinger, S. Smith, S. Wagle, S. Horawalavithana, Nuclearqa: A human-made benchmark for language models for the nuclear domain, arXiv preprint arXiv:2310.10920 (2023).

[17] A. Jain, N. Meenachi, B. Venkatraman, Nukebert: A pre-trained language model for low resource nuclear domain, arXiv preprint arXiv:2003.13821 (2020).

[18] R. V. Guha, D. Brickley, S. Macbeth, Schema.org: Evolution of structured data on the web, Communications of the ACM 59 (2016) 44–51.

[19] A. Both, D. Diefenbach, K. Singh, S. Shekarpour, D. Cherix, C. Lange, Qanary - a methodology for vocabulary-driven open question answering systems, in: Proceedings of the 13th International Conference on The Semantic Web. Latest Advances and New Domains - Volume 9678, Springer-Verlag, Berlin, Heidelberg, 2016, p. 625–641. URL: https://doi.org/10.1007/978-3-319-34129-3_38. doi:10.1007/978-3-319-34129-3_38.

[20] R. Sanderson, P. Ciccarese, H. Van de Sompel, Web annotation data model, W3C Recommendation, 2017. URL: https://www.w3.org/TR/annotation-model/.

[21] R. Hodgson, D. Mekonnen, D. Price, J. Hodges, J. E. Masters, S. J. D. Cox, S. Ray, QUDT: Quantities, units, dimensions and types, 2011. URL: https://qudt.org/. doi:10.25504/FAIRsharing.d3pqw7.

[22] D. Vrandečić, M. Krötzsch, Wikidata: A free collaborative knowledgebase, Communications of the ACM 57 (2014) 78–85.

[23] Publications Office of the European Union, Eurovoc multilingual thesaurus, 2024. URL: https://op.europa.eu/en/web/eu-vocabularies/eurovoc.

[24] T. Pahalage Dona, A. Bulgheroni, L. Fongaro, M. Lissandrini, Proton-qa corpus, 2026. URL: https://doi.org/10.5281/zenodo.22543594. doi:zenodo.22543594.

[25] A. Miles, S. Bechhofer, SKOS Simple Knowledge Organization System Reference, W3C Recommendation, World Wide Web Consortium, United States, 2009.

[26] Google, Notebooklm, https://notebooklm.google.com/, 2026.

[27] S. Es, J. James, L. Espinosa Anke, S. Schockaert, RAGAs: Automated evaluation of retrieval augmented generation, in: N. Aletras, O. De Clercq (Eds.), Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations, Association for Computational Linguistics, St. Julians, Malta, 2024, pp. 150–158. URL: https://aclanthology.org/2024.eacl-demo.16/. doi:10.18653/v1/2024.eacl-demo.16.

[28] J. Saad-Falcon, O. Khattab, C. Potts, M. Zaharia, ARES: An automated evaluation framework for retrieval-augmented generation systems, in: K. Duh, H. Gomez, S. Bethard (Eds.), Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), Association for Computational Linguistics, Mexico City, Mexico, 2024, pp. 338–354. URL: https://aclanthology.org/2024.naacl-long.20/. doi:10.18653/v1/2024.naacl-long.20.